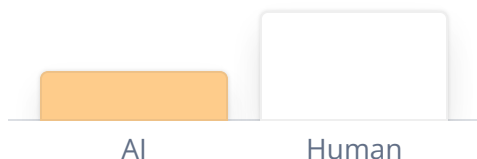




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CULTURE-INDEPENDENT ANALYSIS OF THE BATAK ORAL MICROBIOTA USING HUMAN CONTAMINOME DATA BIO 199.2 – Research in Biology II CHARLES D. PANGUE 2022-09433 BS in Biology (Microbiology) Presented to the Faculty of the Division of Natural Sciences and Mathematics In Partial Fulfillment of the Requirements For the Degree of Bachelor of Science in Biology University of the Philippines TACLOBAN COLLEGE Tacloban City July 2026 ii CULTURE-INDEPENDENT ANALYSIS OF THE BATAK ORAL MICROBIOTA USING HUMAN CONTAMINOME DATA By CHARLES D. PANGUE A research paper submitted to the Division of Natural Sciences and Mathematics University of the Philippines Tacloban College, Tacloban City July 2026 Permission is given for the following people to have access to this research: Available to the general public No Available only after consultation with authors/adviser Yes Available only for those bound by confidentiality agreement Yes Signature of Student: _____ Signature of Research Adviser: _____ iii This is to certify that this research paper “CULTURE-INDEPENDENT ANALYSIS OF THE BATAK ORAL MICROBIOTA USING HUMAN CONTAMINOME DATA”, and submitted by CHARLES D. PANGUE, to fulfill the requirements for the Degree of BACHELOR OF SCIENCE IN BIOLOGY is hereby endorsed. Research Adviser: _____ IRENE L. TAN, PhD Panel Members: _____ FREDERICK C. DELFIN, M.Sc _____ NICCI ROSIETESS E. MADARCOS, M.Sc _____ LENI G. YAP-DEJETO, PhD The Division of Natural Sciences and Mathematics (DNSM) accepts this

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v ABSTRACT This study presents the first culture-independent characterization of the Batak oral microbiota using the human contaminome approach. Unmapped reads derived from whole genome sequencing data generated by the Filipino Genome Research Program (FGRP) were extracted and analyzed using the bioBakery platform, particularly MetaPhlAn v4.1.0 for taxonomic profiling and HUMAnN v3.9 for functional profiling.

The results of this study showed that approximately 72% of the contaminome reads were successfully classified into known or partially known microbial taxa, establishing the reliability of the

contaminome approach as an effective culture-independent technique for analyzing host-associated microbiomes. Taxonomic profiling revealed that the Batak oral microbiome harbors a diverse

microbial community dominated by *Neisseria* and *Prevotella* as core genera together with opportunistic pathogens and uncultured Candidate Phyla Radiation members of the *Saccharibacteria*. Functional profiling further revealed that the community possesses the genetic capabilities for active cellular growth, biosynthetic activity, carbohydrate fermentation, antimicrobial protection through hydrogen peroxide production, and immune-related functions. In addition, interindividual variation was observed to be greater at the species and strain levels than at higher taxonomic ranks, while the overall community structure remained broadly consistent with global oral microbiome patterns.

Overall, the findings of this study establish a baseline description of the Batak oral microbiota and demonstrate the value of contaminome-based culture-independent methods for studying underrepresented indigenous populations.

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INTRODUCTION

The oral microbiota represents one of the most diverse and complex microbial communities in the human body, second only to the gut (Rajasekaran et al. 2024). It is composed of microbial species from Bacteria, Archaea, Eukarya, and viruses (Li et al. 2022; Rajasekaran et al. 2024), with Bacteria dominating the community (Rolstad et al. 2025). This diversity is influenced by various factors such as host genetics, age, diet, oral hygiene, and cultural practices (Demmitt et al. 2017; Mukherjee et al. 2021). Collectively, these microbes play an important role in maintaining oral and systemic health by suppressing opportunistic pathogens (Sedghi et al. 2021) and promoting oral microbial balance, or eubiosis (Santacroce et al. 2023). Disruption of this balance through poor oral hygiene or lifestyle choices may lead to dysbiosis (Degruittola et al. 2016; Rajasekaran et al. 2024), which can contribute to periodontitis and systemic inflammatory conditions (Rajasekaran et al. 2024). Understanding the composition, interactions, and functional roles of oral microbes is therefore crucial to identifying factors that preserve microbial balance and prevent disease development. In studying the oral microbiota, researchers use either culture-dependent or

culture-independent approaches (Perito and Cavalieri 2018). Culture-dependent methods rely on traditional culturing techniques, while culture-independent approaches examine microbial communities directly using genomic data without the need for cultivation (Perito and Cavalieri 2018). In comprehensive microbiome studies, culture-independent methods are increasingly used because they capture both culturable and unculturable microorganisms (McIlwaine et al. 2024).

Traditional culture-independent methods typically involve the collection, extraction, and sequencing of microbial DNA from environmental or biological samples (Su et al. 2012). While these approaches have provided substantial insights into microbial diversity, they require expensive lab equipment, procedures, and associated costs. An alternative strategy involves the use of the “human contaminome”, which refers to unmapped or poorly aligned reads from existing human whole genome sequencing (WGS) data hypothesized to contain non-human, potentially microbial, sequences (Chrisman et al. 2022). By utilizing existing human WGS datasets, this approach offers a cost-effective culture-independent strategy for microbial analysis when such datasets are already available, eliminating the need for additional wet-lab procedures. These unaligned reads can be subjected to taxonomic and functional profiling to infer the potential composition and functional capacity of host-associated microbial communities, although results should be interpreted with caution due to the possible contamination of non-biological artifacts.

To conduct culture-independent analyses advance bioinformatic tools are required. Recent technological developments have produced platforms now capable of processing large metagenomic datasets (Nam et al. 2023). However, many of these platforms are designed to conduct only a single profiling task (Beghini et al. 2021). bioBakery addresses this limitation by providing a multi-analysis environment. Using clade-specific markers, updated microbial reference databases, and advanced filtering tools, bioBakery performs high-resolution taxonomic and functional profiling from metagenomic shotgun data, making it suitable for comprehensive microbiome analyses (Beghini et al. 2021).

Despite advancements in bioinformatics and the growing interest in oral microbiome research, there still remains a large gap, particularly in representations of certain populations or groups. Filipino groups, in particular, remain underrepresented, as little to no data exist that caters to their unique lifestyle, diets, and environments. Moreover, oral microbiome studies are largely biased toward European and highly

industrialized Asian populations due to funding availability and access to resources (Nath et al. 2021). This limitation risk overlooking unique microbial diversity in underrepresented groups, especially among indigenous groups such as the Batak Filipino group of Region IV-B, Palawan. The Batak, a native Negrito group characterized by their genetic distinctness, unique population history, and minimal exposure to urban diets (Scholes et al. 2011), may harbor distinct oral microbial compositions and functional patterns that differ significantly from other Filipino groups and non-Filipino cohorts. Studying this group provides valuable insights into how genetics, lifestyles, culture, and environment shape microbial diversity.

To address this gap, this study utilizes the human contaminome to taxonomically and functionally profile the oral microbiota of the Batak Filipino group of Region IV-B. Coupled with the high-resolution analytical capabilities of bioBakery, this study addresses the underrepresentation of Filipinos in oral microbiome research by providing a cost-effective culture-independent strategy for microbial analysis. Moreover, this methodology aligns with the approach used by the Filipino Genomes Research Program, of which this study is part.

This study aims to taxonomically and functionally profile the oral microbes of the Batak indigenous group using human contaminome data. Specifically, this study aims to achieve the following objectives:

1. To quantify the proportion of classified and unclassified reads in the human contaminome data; 4
2. To determine the taxonomic composition of the Batak oral microbiota across all taxonomic levels;
3. To identify, at genus and species level, the core oral microbes associated with the group;
4. To characterize the inferred functional metabolic potential of the oral microbial community, including pathways related to biosynthesis, metabolism, pathogen protection, and immune modulation; and
5. To establish interindividual variations in taxonomic and functional profiles.

LITERATURE REVIEW

Oral Microbiota and Its Composition

Bacterial Dominance and Diversity

The human oral cavity accommodates a diverse and complex community of microorganisms collectively known as the oral microbiota (Rajasekaran et al. 2024). It

is second only to the gut in diversity (Deo and Deshmukh 2019; Cho et al. 2021) and is regarded as one of the largest microbial communities in the human body (Deo and Deshmukh 2019). Collectively, it harbors microbial species from all three domains of life, Bacteria, Archaea, and Eukarya, with viruses also contributing to community dynamics.

Among these groups, bacteria are the most dominant (Lu et al. 2019) with individuals distributed across diverse phyla. These include representatives from major groups such as Actinobacteria, Bacteroidetes, Firmicutes, Fusobacteria, Proteobacteria, and Spirochaetes, along with less abundant lineages such as Chlamydiae, Chloroflexi, Euryarchaeota, SR1, Synergistetes, Tenericutes, and TM7 (Dewhirst et al. 2010; Rajasekaran et al. 2024).

Given this bacterial dominance, considerable efforts have been made to quantify its diversity. One of the most widely used resources is the Human Oral Microbiome Database (HOMD), a curated database of oral bacterial genome sequences and oral bacterial taxa identified using 16S rRNA (Chen et al. 2010). Earlier versions of this database estimate oral bacterial diversity at about 500 to 700 bacterial species (Chen et al. 2010). However, with advancements in genomic analysis and subsequent update of the database (HOMD v4), this estimate has been expanded to about 834, with 523 of these recognized to be of primarily oral origin (Rolstad et al. 2025). Despite these advancements, many studies still rely on the earlier version of the database as their reference (Deo and Deshmukh 2019; Baker et al. 2023; Rajasekaran et al. 2024). This reliance on outdated information may limit the level of taxonomic precision and underestimate the true diversity of the oral microbiome. Furthermore, large-scale metagenomic surveys have continued to uncover novel and unclassified taxa not represented in curated databases, potentially suggesting that the true diversity might be greater than current estimates (Cha et al. 2025).

While such databases provide insights into oral bacterial diversity, it often overlooks other non-bacterial members of the oral cavity, such as eukaryotes, archaea, and viruses. Although they make up smaller proportions of the community, they participate significantly in host-microbe interaction, metabolic processes, and microbial community regulation.

Eukaryotic, Archaeal, and Viral Composition

Eukaryotes, for instance, exist as microeukaryotes represented mostly by fungal species. Recent studies estimate its diversity to be only 0.1% of the total oral microbial population (Rajasekaran et al. 2024). This low abundance poses a significant challenge for detection, as current taxonomic classification tools often lack the sensitivity to detect low-abundance taxa, leading to their frequent underrepresentation in oral microbiome studies (Achtman and Zhou 2020; Crouch et al. 2024). Nevertheless, despite their low abundance, oral microeukaryotes play an important role in oral microbial dynamics, particularly in shaping fungal community structure. In this context, oral mycobiomes are typically dominated by either *Candida* or *Malassezia* species (Hong et al. 2020; Baker et al. 2023). The dynamics of these two fungal taxa provide insights into oral health and its disruption, with *Candida* generally linked to disease risk and *Malassezia* to oral mycobiome balance (Hong et al. 2020). Hence, the dominance of either *Candida* or *Malassezia* serves as a bioindicator of host conditions. In addition to the eukaryotic component, archaeal members of the oral microbiota have also been documented, although their roles remain less understood. One of the reasons for this is its low prevalence in the oral cavity, which also poses a significant challenge in detection (de Cena et al. 2022). However, efforts have recently been conducted to understand its role, with the recent one being a meta-analysis of existing literature from 1980 to 2020.

Analysis of these literatures reveal that *Methanobrevibacter oralis* appears to be the most common and abundant archaeal taxon in the oral microbiota (Deng et al. 2017; de Cena et al. 2022). The analysis also reveals that it participates significantly in maintaining bacterial balance through the promotion of fermentative bacterial growth (de Cena et al. 2022; Djemai et al. 2022). This ecological role arises from its ability to consume hydrogen (H_2) produced as a by-product of anaerobic bacteria during fermentation, lowering local H_2 concentration and helping fermentative bacteria to continue producing energy (de Cena et al. 2022; Djemai et al. 2022). Without its presence, local H_2 concentrations may accumulate, which makes fermentation energetically unfavorable, affecting the fermentative bacterial population.

This interaction, however, may also harbor detrimental effects, as lower H_2 environments may also favor anaerobic pathogens linked to periodontal diseases (de 8 Cena et al. 2022). Meta-analyses show that archaeal numbers are significantly elevated on hosts with periodontitis and reduced on hosts that have undergone periodontal

treatment (de Cena et al. 2022). These ultimately suggest that oral archaea also serve as indicators of oral health status in addition to their role in maintaining oral bacterial balance.

Similar to archaea, viruses also constitute an important, though often overlooked, component of the oral microbiota. Since the oral cavity is one of the few regions in our body exposed to the external environment, it is not surprising that viruses can also be detected (Thakkar et al. 2021). For instance, bacteriophage, viruses that specifically target bacterial species, are one of the most prevalent viral components of the oral cavity (Rajasekaran et al. 2024). They are not transient invaders, as most viruses are, rather they are considered normal residents of the oral cavity, forming part of the stable oral ecosystem in healthy individuals (Pride et al. 2012; Guo et al. 2024). Studies on viral interactions in the oral microbiota suggest that oral bacteriophages play a significant role in maintaining bacterial balance and dynamics by preventing the overgrowth of bacterial species and facilitating horizontal gene transfer (Jalili et al. 2025). Such interactions help sustain oral microbial balance or eubiosis (Santacroce et al. 2023).

Since the oral cavity serves as the primary entry for microbial pathogens, it can also be home to various eukaryotic viruses. Human herpesviruses are considered the most commonly detected eukaryotic viruses in the oral cavity (Thakkar et al. 2021). It is highly prevalent in humans, with 90% of the global population infected by one or more of its members (Thakkar et al. 2021). Furthermore, the oral cavity serves as the main entry point for herpesviruses, contributing to their high prevalence in the oral cavity (Thakkar et al. 2021).

Understanding the full diversity of these viral members, however, depends on the methods of detection employed. Historically, viral detection in the oral cavity has relied mostly on conventional techniques such as viral culture, PCR, serological assays, and transmission electron microscopy (Roingear 2008; Lappalainen and Niesters 2021; Cassedy et al. 2021). However, these methods carry a significant limitation as most viruses cannot be grown easily or cannot be grown in vitro at all (Lappalainen and Niesters 2021). The emergence of metagenomics and next-generation sequencing has since revolutionized viral detection in the oral cavity, making it suitable for profiling bacteriophages and discovering novel human viruses (Parras-Moltó and López-Bueno 2018). Nevertheless, this approach also has its own limitations, as metagenomic detection of viruses is heavily constrained by the reference database used for

bioinformatic classification. In fact, there are cases that 99% of putative viral sequences in a sample cannot be taxonomically classified because of their high diversity and limited representation in existing databases (Santiago-Rodriguez and Hollister 2022).

These detection limitations are not just unique to viruses but also extend to other low-abundance microbial members of the oral cavity, including eukaryotes and archaea, as previously discussed. Collectively, these limitations in detection highlight the need for expanded and curated databases to include viral sequences, improved bioinformatic processes to enhance sensitivity on low-abundance taxa, and the development of new strategies for database-independent metagenomic characterization approaches of the oral microbiome.

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Oral Microbiota and Its Link to Oral and Systemic Health

Oral health studies reveal that a healthy oral cavity is supported by a balanced microbial community by allowing beneficial microbes, particularly bacterial species, to thrive (Rajasekaran et al. 2024). These beneficial bacterial species, also known as commensals, play a protective role in the oral cavity by preventing the growth of opportunistic pathogens through a mechanism called resistance to colonization (Sedghi et al. 2021). In this process, commensals directly antagonize oral pathogens by releasing antagonistic chemicals that inhibit the growth of opportunistic microbes, thereby outcompeting them in integrating and colonizing the oral cavity (Sedghi et al. 2021; Rajasekaran et al. 2024). Thus, this literature suggests that bacteria, fungi, archaea, and viruses together form a complex and interrelated oral ecosystem whose interactions with one another are crucial for maintaining oral microbial balance. Maintaining this balance is essential as oral health relies heavily on the stability of this community. However, this delicate balance can be disrupted by poor oral hygiene, dietary habits, medical conditions, medications, and lifestyle choices (Rajasekaran et al. 2024). This disruption, also known as oral dysbiosis, can lead to the development of oral health problems such as dental caries, periodontal diseases, halitosis, and oral candidiasis (Rajasekaran et al. 2024). If the natural balance is not restored, these oral health issues may extend beyond the oral cavity, as emerging studies on oral microbiome and oral health suggest potential links between oral dysbiosis and systemic diseases, including cardiovascular disease, diabetes, and cognitive disorders (Thomas et al. 2021; Rajasekaran et al. 2024).

A key characteristic of oral dysbiosis is the overgrowth of harmful pathogens and the decline of beneficial microbes (Degruttola et al. 2016; Rajasekaran et al. 2024). 11 This imbalance causes pathogens to proliferate, damaging tissues and compromising natural processes linked to tissue repair, further magnifying systemic health problems (Rajasekaran et al. 2024). Therefore, the oral microbiota emerges not just as a critical factor in maintaining oral health but also in influencing systemic health conditions through host-microbe interactions. Understanding these interactions would require a comprehensive understanding of the microbial community through detailed characterization of its diversity and relationships. Phylogenetic analysis serves as an essential tool for acquiring this understanding, as it provides a framework for examining microbial diversity, providing insights into their relationships and potential roles within the oral microbiome.

Core Microbiome

Broadly speaking, the core microbiome refers to any set of microbial taxa that are characteristic of a host or environment (Neu et al. 2021). For it to be considered a characteristic of the environment, these microbial taxa should be consistently present across a population (Sharon et al. 2022). Despite this definition, the concept of the core microbiome still remains ambiguous as there are no universally accepted criteria used to define it. Different studies define the core microbiome differently depending on the objectives of the research study.

There are several ways research studies define the core microbiome. Some studies used the occurrence- or prevalence-based approach which identifies taxa as members of the core if it is consistently detected across individuals in a population regardless of its relative abundance (Neu et al. 2021). Other research studies, on the other hand, use abundance-based approach in which microbial taxa that falls within a 12 specific relative abundance thresholds are considered core (Neu et al. 2021). There are some cases as well that both the occurrence- and abundance-based methods are combined. In this method, the core is defined as the microbial taxa that are consistently present within a population with a relative abundance high enough for it to be detected (Neu et al. 2021). This combined method is often viewed as more reliable as it provides a more conservative description of the core microbiome by identifying microbial taxa

that are both widespread and ecologically important within the community (Neu et al. 2021).

In occurrence-abundance approach, occurrence and abundance thresholds are determined by the researcher depending on the objectives of the research study. In many studies, occurrence thresholds ranged between 50 to 100% (Neu et al. 2021) while abundance thresholds ranged between 0.001 (Antwis et al. 2018) to 4.5% (Leis and Costa 2019). Although this approach is considered a reliable method for identifying the core microbiome because it accounts for taxa that are both consistently present and relatively abundant, several nuances and assumptions must still be considered. Differences in phylogeny, ecological roles, spatial distribution, and sampling size may influence abundance–occurrence relationships among microbial taxa, which can affect the identification and interpretation of the core microbiome (Neu et al. 2021). Therefore, researchers must carefully select and justify the thresholds and criteria used in defining the core microbiome to ensure that the results are biologically meaningful and appropriate for the study population.

The Batak Indigenous Group and Oral Microbiota of Indigenous Populations

Indigenous populations possess a distinct oral microbiota as their traditional lifestyle practices and eating habits as well as their limited contact with industrialized environments create unique microbial ecosystems (Dobon et al. 2023). The Batak Indigenous Group of the Philippines is one of the remaining indigenous communities that still practice aspects of traditional subsistence lifestyles. Their dietary patterns follow a pattern similar to that of other hunter–gatherer and semi-hunter–gatherer groups as they consume both minimally processed foods and plant-based resources (Novellino 2015), which create unique oral microbial communities. Research on indigenous communities in the Philippines, particularly that of the Agta hunter–gatherers from the Philippines showed that dietary and lifestyle factors determine the composition of their oral microbiome (Dobon et al. 2023). The transition from

traditional protein- and fiber-rich diets to rice-rich agricultural diets led to changes in oral microbial species distribution (Dobon et al. 2023). The traditional eating patterns of indigenous and hunter-gatherer groups lead to their body systems favoring *Neisseria* and *Prevotella* (Dobon et al. 2023). The presence of *Prevotella* in non-Western diets often containing fiber explains its association with this dietary pattern, while *Neisseria* serve as a basic component of human oral microbiota found in non-industrialized societies (Dobon et al. 2023). These findings suggest that the predominance of these genera in indigenous populations may reflect adaptations to traditional lifestyles and dietary practices rather than dysbiosis.

Phylogenetic and Taxonomic Analysis of Oral Microbes

Phylogenetic analysis has long been an established method in understanding community structure and evolutionary relationships among organisms, even dating back to Darwin's time (Vitt and Caldwell 2009). Historically, before the advent of DNA sequencing technologies, phylogenetic analysis relied mostly on traditional taxonomic features such as morphology and observable traits to infer phylogenetic trees (Zou et al. 2024).

However, with advancements in biotechnology and the rise of sequencing tools, molecular phylogenetics became the standard (Zou et al. 2024). This was a major advancement since molecular phylogenetics provides more comprehensive, quantifiable, and unambiguous data in comparison to the limited information that morphology-based phylogenetics can provide (Brown 2002). However, this does not mean that morphology-based phylogenetics is less valuable than its molecular counterpart, as both have their own advantages and can even be used together to provide more information (Keating et al. 2023). Nevertheless, researchers often opt for molecular phylogenetics due to its advantages.

Two common methods are used in molecular phylogenetics: distance-based methods and character-based methods. Distance-based methods infer phylogenetic

trees by converting a feature matrix to a distance matrix to represent evolutionary distances and combining them using clustering algorithms (Zou et al. 2024). On the other hand, character-based methods use optimality criteria such as parsimony, minimum evolution, or maximum likelihood to infer phylogenetic trees (Zou et al. 2024). Both of these methods are largely computer-based, relying on computer algorithms to infer the optimal phylogenetic tree. Moreover, each method also has its own advantages and limitations, and a deep understanding of these factors is required in choosing the best methods for a particular analysis (Zou et al. 2024).

In the context of oral microbiota, recent studies used molecular phylogenetics to understand the diversity and evolutionary relationships among oral microbes, with most of these studies leveraging metagenomic data. In these studies, they often used 16S-based phylogenies to provide broad surveys of community composition and diversity or whole genome approaches via genome-resolved metagenomics for higher resolution (Manghi et al. 2024). Whole genome approaches typically allow for a much deeper analysis with the ability to distinguish between closely related strains or linked metabolic potential to phylogenetic position compared to 16S analysis (Utter et al. 2020; Li et al. 2023; Manghi et al. 2024).

Using these approaches, researchers are now able to catalog key microbial taxa present in the oral microbiota and their relationships with one another. Members of the Actinomyces, Streptococcus, Fusobacterium, the Cytophaga-Flavobacterium-Bacteroides (CFB) cluster, Tannerella, and periodontal pathogens such as Porphyromonas and Treponema are among the microbial taxa frequently documented in recent phylogenetic studies (Chen et al. 2018; Willis and Gabaldón 2020; Velsko and Warinner 2025).

Further phylogenetic analyses also show that some of these groups form well-defined clades sharing similar conserved functional profiles. For instance, Streptococcus strains (e.g., *S. mitis*, *S. sanguinis*, and *S. mutans*) sometimes form distinct phylogenetic clades having similar saccharolytic pathways and site tropisms (Takahashi 2015; Abranches et al. 2018; Velsko and Warinner 2025). Recognizing this pattern may help explain why some closely related strains share similar functional profiles, ultimately bridging phylogenetics and functional analysis. In this sense, phylogenetic analysis provides a foundation for predicting metabolic potentials, roles, or behaviors of microbes, making functional profiling an extension of phylogenetics.

Functional Profiling of Oral Microbes

Functional profiling focuses on identifying the metabolic activities and roles of microorganisms within the oral community. Specifically, it is a method by which metabolic capabilities of microbial communities are characterized based on their genetic data (Franzosa et al. 2018). Functional profiling gives insights into what the microbes are able to do. It does not give insights into what the microbes are currently doing since this constitutes functional activity. Functional profiles may include metabolic pathways (e.g., sugar fermentation, protein degradation, or vitamin synthesis), antibiotic resistance (i.e., presence of genes that confer resistance to antibiotics), or host-interaction (e.g., adhesion, immune evasion, or toxin production that allows microbes to colonize or cause disease) within the oral cavity (Franzosa et al. 2018; Casimiro-Soriguer et al. 2019).

To uncover these profiles, researchers utilize metaomic data to identify genes and pathways associated with functions such as metabolism, antibiotic resistance, and host interaction (Beghini et al. 2021; Hera et al. 2024). Traditionally, functional profiles use metatranscriptomic data to accurately capture the actual activities of microbes within a community. However, implementation of such analysis is often expensive and tedious due to the nature of the equipment used for collecting and analyzing expression data (Shakya et al. 2019; Jovel et al. 2022). Thus, researchers often resort to 17 metagenomic analysis as it is more cost-effective than metatranscriptomic analysis (Jovel et al. 2022).

However, using metagenomic data for functional analysis is limited only to the analysis of potential functions, not the actual activity (Tanca et al. 2017; Jansson and Hofmockel 2018). Direct evidence of actual microbial function would require analysis of expressed genes in the form of metatranscriptomic, metaproteomic, or metabolomic data (Aguilar-Pulido et al. 2016; Galloway-Peña and Hanson 2020). Nevertheless, even without expression data, metagenomic data still provide valuable information about the functional potential of microbes in the community (Heidrich et al. 2022). In fact, metagenomic functional profiling has been an established method in many microbiome research studies. Studies that use this type of analysis serve as baseline studies for understanding the functional potential of microbes and for drawing valuable hypotheses for further testing (Heidrich et al. 2022). Understanding this limitation avoids drawing

false conclusions on functional profiles without supporting them with expression data (Heidrich et al. 2022). In the context of oral microbiome studies, profiling of functional potential is especially significant as it provides baseline data for further functional analysis. Moreover, it also provides valuable information for inferring links between oral microbial processes and host physiology, including nutrient metabolism and overall health (Rajasekaran et al. 2024).

Building on this, analysis of functional profiles has revealed functional adaptations of oral microbes in unique ecological and cultural contexts. A spatial analysis of Neisseriaceae species, for instance, reveals that they occupy distinct niches in the oral cavity with specific adaptations either for nitrogen metabolism, lysine degradation, and galactose metabolism, depending on their location (Giacomini et al. 2025). In the cultural context, on the other hand, cultural transitions, as demonstrated by Agta hunter-gatherers, where dietary shifts from fish-meat to rice-rich diets are apparent, influence microbiome composition and thus their functional profiles (Dobon et al. 2023). These studies show applications of functional profiling and how it can uncover unique functional adaptations of microbes in distinct environments or cultures.

Methods and Tools for Studying Microbiomes

Throughout the years, advances in metagenomic technologies and bioinformatic tools, accompanied by the increasing use of culture-independent analysis, have made it possible for studies on microbial communities to be conducted (Nam et al. 2023). Culture-independent analysis, in particular, enables researchers to profile all microbes in a specific community due to its ability to detect both culturable and unculturable microorganisms in contrast with its culture-dependent counterpart (McIlwaine et al. 2024). This analysis requires extensive sequencing data, which recent advancements in next-generation sequencing technologies are now capable of generating at high-throughput. Currently, two main methods are used in culture-independent analysis of microbiomes using high-throughput sequencing data: marker gene analysis and shotgun metagenomics (Pérez-Cobas et al. 2020). Each of these methods differs significantly in focus and depth of analysis.

Marker Gene Analysis

Marker genes refer to highly conserved regions within microbial genomes that are targeted during amplification and subsequently sequenced for analysis. Common 19 examples include the 16S ribosomal RNA (rRNA) gene in bacteria, the internal transcribed spacer (ITS) region in fungi, and the 18S rRNA gene in eukaryotes, along with other less commonly used marker genes (Pérez-Cobas et al. 2020; Galloway-Peña and Hanson 2020). Although these genes are highly conserved, they have accumulated sufficient variations over time, serving as unique molecular barcodes that enable the identification of taxonomic groups and the assessment of microbial diversity and community composition (Pérez-Cobas et al. 2020; Galloway-Peña and Hanson 2020).

Marker gene analysis offers several advantages, particularly for studies involving large numbers of samples. It is generally faster to process, simpler to analyze, and less expensive than shotgun whole-genome metagenomic approaches (Pérez-Cobas et al. 2020). These characteristics make it an efficient and widely used method for profiling microbial communities.

However, this type of analysis also has limitations. One challenge lies in the presence of hypervariable regions within marker genes, which can change rapidly over short evolutionary timescales and may even vary within a single genome. This variability can reduce the accuracy of taxonomic identification (Galloway-Peña and Hanson 2020). Another limitation is the occurrence of shared or identical gene sequences among distinct taxa. For instance, bacterial genera such as *Escherichia* and *Shigella* possess nearly identical 16S rRNA sequences, complicating their differentiation and accurate classification (Větrovský and Baldrian 2013; Galloway-Peña and Hanson 2020).

Because marker gene sequencing primarily targets specific genomic regions, it can only provide microbial taxonomy (Patel et al. 2023). Hence, it is well-suited for describing diversity and composition, but cannot provide detailed functional profiles or 20 activity. Taxonomic assignments in marker gene analysis are typically performed using machine learning-based classification algorithms that compare the obtained sequences to reference databases, enabling the identification and categorization of organisms present in a given sample (Galloway-Peña and Hanson 2020).

Shotgun Metagenomics

In contrast to marker gene analysis, shotgun metagenomics employs an untargeted sequencing approach that aims to capture all microbial genomes present within a sample. This strategy enables the simultaneous analysis of diverse microbial groups, including bacteria, archaea, eukaryotes, and viruses, rather than focusing on a single taxonomic marker (Qin et al. 2010; Galloway-Peña and Hanson 2020). By sequencing the entire genomic content, shotgun metagenomics allows not only the characterization of community composition but also the exploration of functional profiles, offering deeper insights into the metabolic potential and ecological roles of microorganisms (Qin et al. 2010; Galloway-Peña and Hanson 2020; Patel et al. 2023). Moreover, it also captures taxonomic resolution at the species or strain level.

Shotgun metagenomic analysis relies heavily on assembly-based methods to reconstruct genomes from the sequenced fragments. These assemblies can be performed de novo, reference-based, or through hybrid approaches that combine both strategies (Galloway-Peña and Hanson 2020). Currently, there is no universal consensus on which assembly method is most effective (Galloway-Peña and Hanson 2020), as the optimal choice depends largely on the objectives, sample complexity, and data quality of a given study. 21

Despite its comprehensiveness, shotgun metagenomics also presents several limitations. Its reliance on large datasets and complex assembly algorithms makes it computationally intensive. Additionally, assembling highly diverse microbial communities remains a technical challenge, which can impact the accuracy of genome reconstruction and downstream analyses (Galloway-Peña and Hanson 2020). Nonetheless, shotgun metagenomics provides a powerful framework for simultaneous taxonomic and functional profiling, making it a valuable complement to marker gene-based studies.

Bioinformatic Analysis Tools and bioBakery

Conducting such methods requires the use of bioinformatic tools or software that are capable of processing and analyzing marker-gene sequence or shotgun metagenomic data. Tools such as Kraken2 for taxonomic classification (Wood et al.

2019), eggNOG-mapper for functional annotation (Cantalapiedra et al. 2021), and Bayesian Evolutionary Analysis Sampling Trees (BEAST) for phylogenetic analysis (Drummond and Rambaut 2007) and other similar tools are now accessible and have made such analyses more efficient. However, many of these tools are designed only to accommodate a single, specific type of analysis (Beghini et al. 2021), which poses a challenge for studies that require multiple analytical approaches. In such cases, researchers must install, configure, and integrate several different software or tools, which can be computationally intensive and memory-demanding.

Addressing this limitation is the development of platforms that integrate multiple analytical approaches into a single environment. Examples of this include QIIME 2, which provides a modular framework for marker gene data processing, 22 taxonomic assignment, and diversity analysis (Bolyen et al. 2019), and MEGAN, which allows both taxonomic and functional analysis of metagenomic datasets within one platform (Huson et al. 2007). Although these platforms provide an integrated approach to microbiome analysis, they are limited to amplicon-based data or read-alignment-based classification, respectively (Beghini et al. 2021). This limits their interpretative depth and resolution.

To further address this, a new platform, known as bioBakery, was developed.

This platform builds upon the limitations of earlier integrated platforms, aiming to deepen the resolution and accuracy of microbiome studies. Similar to the previously mentioned platforms, bioBakery provides a comprehensive tool suite for taxonomic, functional, and strain-level profiling of metagenomic, metatranscriptomic, and other microbial community data. Specifically, the tool uses MetaPhlAn for taxonomic classification and HUMAnN for functional profiling, all of which use an updated ChocoPhlAn database as the primary microbial reference (Beghini et al. 2021). In contrast to other comparable analysis tools, such as those previously mentioned, bioBakery is able to analyze whole metagenomic shotgun data and uses clade-specific markers, allowing for a more accurate functional profiling and high-resolution taxonomic analysis reaching until the species level (Beghini et al. 2021).

MetaPhlAn Taxonomic Profiling

MetaPhlAn is a computational tool integrated in the bioBakery platform

designed for the taxonomic identification of microbes comprising a specific microbial community. Unlike traditional approaches that rely on read assembly or full-genome alignment for taxonomic assignment, MetaPhlAn uses a set of clade-specific marker 23 gene sequences, which are unique sequences found only on specific microbial taxa (Beghini et al. 2021). These sequences serve as taxonomic barcodes that allow the tool to accurately identify and estimate relative abundances of microbial taxa such as bacteria, archaea, eukaryotes, and viruses that might be present in a sample.

MetaPhlAn uses an expanded reference database, ChocoPhlAn, which contains millions of unique markers derived from over 100,00 reference microbial genomes (Beghini et al. 2021). Raw and pre-processed sequencing reads are directly mapped against this marker-gene database using a fast-alignment algorithm, Bowtie2, that is also integrated within the platform (Beghini et al. 2021). The use of this database bypasses the need for genome assembly, reducing computational time and resource requirements. Moreover, this marker-based strategy eliminates false positives and increases resolution of taxonomic assignments, reaching the species or strain level. This tool is often used in shotgun metagenomic studies as a taxonomic component of the bioBakery tool suite, providing improved accuracy, sensitivity, and coverage compared to its previous versions.

HUMAnN Functional Profiling

HUMAnN, which stands for The HMP Unified Metabolic Analysis Network, is the functional analysis component of the bioBakery. It allows for the identification of microbial activities within a specific community, providing insights into their functional potential. Specifically, HUMAnN maps sequencing reads to species-specific reference pangenomes derived from the ChocoPhlAn database for gene family assignment (Beghini et al. 2021). Reads that do not align with the reference database are further aligned to a more general protein database, primarily UniRef90. This 24 analysis builds gene families, providing insight into the various genes that may contribute to specific biochemical processes or pathways (Beghini et al. 2021). Moreover, HUMAnN also calculates the relative abundances of each gene family in the community (Beghini et al. 2021).

After identifying the gene families present, these are then mapped to metabolic

pathways using the MetaCyc database. This analysis reconstructs metabolic pathways that may be present based on the metagenomic data and estimates both pathway abundance and coverage (Beghini et al. 2021). Pathway abundance refers to the amount of genetic material contributing to a specific pathway, while pathway coverage refers to whether the pathway is fully encoded by the community. Together, this reconstructs the community's functional potential, or the biochemical processes that the microbiome could carry out (Beghini et al. 2021). HUMAnN complements MetaPhlAn in the bioBakery platform as an extension to taxonomic analysis, providing information on the biological roles and activities of the identified microbes in a specific community.

The Human Contaminome for Microbiome Analysis

In addition to conventional microbiome analysis methods, new approaches in microbiome analysis have also emerged. Traditional culture-independent analysis of microbiomes usually involves the direct collection and sequencing of microbial DNA from environmental or biological samples. In oral microbiome studies, researchers usually collect saliva, dental plaque, or swab samples, which are then subjected to either metagenomic shotgun sequencing or 16S rRNA amplicon-based sequencing. While these approaches provide comprehensive and accurate insights into microbial composition, they are costly, time-consuming, and resource-intensive. 25

An emerging and cost-effective approach uses the “human contaminome,” which refers to the unmapped or poorly aligned reads in human whole genome sequencing data (Chrisman et al. 2022). These reads, initially regarded as sequencing noise, were found to contain microbial signatures that can be utilized for analysis (Chrisman et al. 2022). In practice, WGS data are initially aligned against the human reference genome, and reads that did not align are extracted and then mapped against microbial reference genomes to identify the presence of bacterial, viral, and other microbial taxa (Chrisman et al. 2022). This approach allows researchers to recover microbial information from existing WGS data without additional laboratory requirements, making it a cost-effective, culture-independent, and accessible approach. However, despite its promising potential as an alternative and perhaps a novel approach for microbiome analysis, the human contaminome approach also presents several limitations. One such limitation is the low abundance of microbial reads in

WGS data. This limitation reduces the sensitivity of detecting rare taxa or limits functional profiling. Moreover, potential laboratory or reagent contamination may also introduce false positives. Hence, care must be taken to eliminate these unwanted contaminations. An advanced and efficient analysis tool is needed so that these limitations can be bypassed, and many tools are already available.

Currently, there are limited studies that use this approach for microbiome analysis. However, it has been successfully applied in oral and systemic microbiome studies showing its ability for retrospective, population-scale analyses (Chrisman et al. 2022). Using this approach provides a novel and innovative strategy complementary to conventional microbiome methodologies only if existing whole genome sequence dataset is available. 26

Filipino Genomes Research Program (FGRP)

The Filipino Genomes Research Program (FGRP) is a national research initiative that aims to represent Filipino groups in global genomic research (DOST-PCHRD 2022). This national initiative is composed of three projects which includes Project 1: Filipino Forensic Genomics, which aims to generate a Filipino-specific database for forensic analysis, Project 2: Filipino Genomes: History, Evolution, Origins, and Applications which aims to create a Filipino-specific whole-genome and sample resource for various applications, and Project 3: Filipino Genome Regions to Help Resolve Child Sexual Abuse Cases which aims to validate process of using biological samples from child victims of sexual abuse to resolve sexual abuse cases (DOST-PCHRD 2022).

Specifically, this research study is aligned with the Project 2 of the initiative.

Currently, the Project has already completed its Phase 1 of implementation while Phase 2 is already underway. The FGRP Project 2 Phase 1 was funded by the Philippine Council for Health Research and Development (PCHRD Project Number: FP190024), Department of Science and Technology (DOST) from 2019 to 2025. Its Phase 2 implementation, on the other hand, is considered by DOST-PCHRD for priority funding in 2026. The project is implemented by the DNA Analysis Laboratory (DAL), Natural Sciences Research Institute (NSRI), University of the Philippines Diliman (UPD), with research ethics clearance from the University of the Philippines Manila, Research Ethics Board (UMPRESB Code: 2019-0346-01).

Implementation of the Phase 1 of the project generated whole genome sequence dataset from 42 Filipino groups, including that of the Batak indigenous group of

Palawan Region IVB. The generation of this whole genome sequence dataset was done 27 through extraction and sequencing of DNA derived from saliva samples. The existence of this project and the whole genome sequence dataset they generated provides a valuable resource that can be utilized for human-contaminome analysis of the oral microbiota especially in indigenous populations.

Global Trend and Bias in Oral Microbiome Research

Despite advancements in analytical methods and tools, there still remains a large gap in oral microbiome studies, particularly on representations of certain populations or groups, such as Filipinos. Currently, oral microbiome studies in the Filipino population remain limited. More broadly, Asian populations, particularly from less industrialized countries, such as the Philippines, are largely underrepresented in global oral microbiome studies, as most existing literature is concentrated on Western nations and highly industrialized Asian countries like China, where greater funding and resources are available to conduct such studies (Nath et al. 2021). This Eurocentric focus and bias on highly industrialized Asian countries is reflected not only in the available research but also in the microbial reference data commonly used in microbiome analyses, which largely represent European and North American populations (Nath et al. 2021). Such limitations risk overlooking microbial diversity unique to Filipinos and other Asian groups.

Overall, the oral microbiota serves as one of the most important microbiomes in the human body due to its diversity and its implications for oral and systemic health. Understanding its composition and dynamics is crucial for uncovering how microbial interactions and evolutionary relationships influence oral eubiosis and disease susceptibility. Achieving this understanding relies on phylogenetic analysis and 28 functional profiling, which can now be performed using advanced approaches such as bioBakery and the utilization of the human contaminome. Despite these advancements, significant gaps remain, particularly among underrepresented populations. Studying these groups may reveal novel microbial taxa, unique functional adaptations, and insights into the role of the oral microbiome across diverse human populations.

METHODOLOGY

Data Source, Sample Selection, and Ethics Clearance

The primary data used in this study were derived from a Filipino whole genome sequence dataset extracted from saliva samples generated by the Filipino Genomes Research Program (FGRP) Project 2 – Filipino Genomes: History, Evolution, Origins, and Applications Phase 1. This dataset includes genomic data from 451 individuals, composed of 34 parent-offspring trios (68 parents and 34 offspring) and 349 unrelated individuals sampled from 42 Filipino groups.

From these 42 groups, one dataset was randomly selected for analysis, specifically, the dataset corresponding to the Batak group of Palawan (Region IV-B). This dataset contains genomic data from 10 individuals representing the group. The use of this data is covered under the ethics clearance of the University of the Philippines Diliman, Natural Science Research Institute, DNA Analysis Laboratory, Filipino Genomes Research Program, as approved by the University of the Philippines Manila Research Ethics Board (UPMREB: 2019-0346-01).

Extraction of Unmapped Reads

As the starting data for analysis, the study utilized whole-genome sequencing reads from the 10 individuals of the Batak group stored in a BAM (Binary Alignment/Map) file format. These files contained sequencing reads that have been aligned to the human reference genome, as well as any reads that remained unmapped. The unmapped reads were the primary focus of this study and were extracted, sorted, and converted into paired-end FASTQ format using SAMtools v1.10. The resulting FASTQ files, corresponding to the forward and reverse reads of each individual, served as the primary input for taxonomic and functional profiling.

Taxonomic Profiling, Classified and Unclassified Reads Estimation, and Taxon Counts per Taxonomic Level

Taxonomic profiling and relative abundance estimation of the microbial community for each sample were performed using MetaPhlAn v4.1.0. The analysis was conducted using default parameters, with the addition of `--add_viruses` and `--unclassified_estimation` options to enable detection of viral taxa, if present, and to estimate the proportion of classified and unclassified reads. In the context of MetaPhlAn taxonomic profiling, classified reads refer to sequencing reads that were successfully assigned to known microbial taxa based on the reference database, whereas unclassified reads are those that did not match any reference sequences in the database and therefore could not be assigned to a specific microbial taxon. The proportions of classified and unclassified reads for each sample were subsequently noted and consolidated into a single table, in which the mean and standard deviation were calculated to determine the average proportions across all samples as well as to assess variability between samples.

The MetaPhlAn marker gene database

`mpa_vJun23_CHOCOPhlAnSGB_202403` was used for taxonomic assignment. This database version was selected to ensure compatibility with downstream functional profiling using HUMAnN v3.9, facilitating seamless integration of taxonomic and functional results.

The resulting taxonomic profiles from each sample were then merged into a single table containing all samples as columns and the identified microbial taxon as rows using the `merge_metaphlan_tables.py` utility script. The merged profile was then stratified by taxonomic rank from domain to strain level using the `awk` command in Linux to generate separate abundance tables for each level, which were used for downstream analysis in R, particularly at the genus and species level.

Following stratification, the number of unique microbial taxa detected at each taxonomic level was counted and consolidated. This process was performed both per sample, to determine the number of taxa detected at each taxonomic level for each sample, and across the community, to determine the overall number of unique taxa detected at each taxonomic level across all samples. The mean and standard deviation were subsequently calculated for the per-sample table to determine the average count of unique taxa detected and to assess between-sample variability.

Identification of Core Microbiome at Genus and Species Level

Following taxonomic profiling, the core microbiome was then identified at both the genus and species level. Identification of the core microbiome was done using RStudio. Before analysis, the stratified MetaPhlAn tables for both genus and species level, containing all samples, were normalized into relative abundance based on classified reads to ensure comparability across samples. Core microbiome analysis was then performed using a combined prevalence and abundance threshold, defined as $\geq 80\%$ prevalence across samples and $\geq 0.01\%$ relative abundance. Taxa meeting these criteria were considered members of the core microbiome. The identified core microbiome was then filtered to include only the top 15 taxa based on relative abundance, and the rest were pooled and designated as "Others". The results were then visualized using a stacked bar plot showing the distribution of the core microbiome across the samples.

Functional Profiling

Before functional profiling, paired-end reads in FASTQ file format for each

sample were concatenated into a single FASTQ file using the cat command in Linux. This combined file was used as input for functional analysis. The generated MetaPhlAn taxonomic profile per sample was also used as input, in addition to the concatenated paired-end FASTQ files. Functional profiling of the microbial community was then performed using HUMAnN v3.9. This version was selected due to its compatibility with taxonomic profiles generated using MetaPhlAn v4.1.0. The analysis was conducted using default parameters.

For nucleotide-level alignment, the ChocoPhlAn database, full_chocophlan.v201901_v31, was used as the reference database as required by HUMAnN v3.9. For protein-level alignment, the UniRef90 EC-filtered database was utilized to reduce computational time, as it is smaller in size compared to the full UniRef90 database while still retaining functionally informative sequences.

The functional profiling step generated three outputs per sample: gene family abundance, pathway abundance, and pathway coverage. These sample-level tables were subsequently merged into a single table containing all the samples using the humann_join_tables utility script, resulting in one consolidated table for each data type. The resulting datasets represent the functional composition across all samples at the levels of gene family abundance, pathway abundance, and pathway coverage. 33

Normalization of Functional Outputs

Before downstream R analysis, the combined functional outputs at the levels of gene family abundance and pathway abundance were normalized to eliminate the effects of sequencing depth as well as to make the samples comparable to each other during analysis. The normalization was done using the humann_renorm_table utility script, and the outputs were normalized into counts per million (CPM). The pathway coverage output was not further normalized, as its values are already scaled between 0 and 1 and are already comparable between samples.

Regrouping Gene Families and Renaming to Human-readable Format

The normalized merged gene family abundance table in UniRef90 labels was regrouped into MetaCyc reaction-level profiles using the humann_regroup_table utility script with the uniref90_rxn mapping. This step aggregated gene families contributing to the same biochemical reactions to generate function-level profiles. Subsequently,

reaction identifiers were converted into human-readable names using the `humann_rename_table` utility with the metacyc-rxn naming scheme to facilitate easier interpretation of the results.

Splitting Stratified and Unstratified Tables

The normalized and annotated merged functional outputs, including gene family abundance, pathway abundance, and pathway coverage, were further divided into stratified and unstratified tables using the `humann_split_stratified_table` utility script. This step was done to account for different levels of analysis. Stratified tables were used 34 to assess species-specific contributions to each functional feature, whereas unstratified tables represented the overall community functional profile.

Analysis of Most Abundant Pathways, Their Completeness, and Species Contribution

Identification of the most abundant pathways was done using the unstratified pathway abundance table in RStudio. The identification was done by taking the mean abundance of each pathway across the sample and then sorting it in descending order. The top 15 pathways with the highest mean abundance were selected for further analysis.

The corresponding unstratified pathway coverage table was then used to assess the completeness of the top 15 pathways. Coverage thresholds were applied to classify pathways as complete, partial, or incomplete. Pathways with coverage $\geq 80\%$ were considered complete, those with coverage between 50% and 80% were considered partial, and those with coverage $< 50\%$ were considered incomplete.

The stratified pathway abundance table was subsequently used to determine species-level contributions to the top 15 pathways. The results were visualized using a lollipop plot and a heatmap to represent pathway abundance, completeness, and species contributions, respectively.

Analysis of Oral Related Functions

The community was assessed based on its ability to perform three oral-related functions: metabolism of carbohydrates and pH balance, protection against pathogens, and immune system-related functions. All of these analyses were done using RStudio, utilizing all three HUMAnN functional outputs.

For the carbohydrate metabolism ability of the community, the analysis was conducted by filtering only the pathways that correspond to carbohydrate metabolism in the unstratified pathway abundance table and matching them to the species that contribute to those pathways in the stratified pathway abundance table. The identified pathways and their contributing species were then mapped against each other and visualized using a heatmap to show the distribution of species per pathway and which species contribute to each of the identified pathways. In addition, the pathway coverage table was also used in order to determine the completeness of the identified carbon fermentation pathways. They were classified on three criteria based on their mean coverage score: complete ($\geq 80\%$), partially complete (between 50% and 80%), and incomplete ($< 50\%$). On the other hand, the acid-alkali balance profile of the microbial community was assessed by identifying pathways associated with acid-producing carbohydrate metabolism and alkali-producing protein and amino acid metabolism from the stratified pathway abundance table. For each species, the abundances of acid-associated and alkali-associated pathways were summed separately, and an acid-alkali balance score was calculated using the formula:

Balance Score= Alkali CPM–Acid CPM

Alkali CPM+Acid CPM Eq 1. 36

where negative values indicate net acid-producing activity and positive values indicate net alkali-producing activity. The top 20 species based on total pathway abundance were selected for visualization using a diverging bar plot.

The ability of the microbial community to provide protection against pathogens

was assessed by determining the hydrogen peroxide (H₂O₂)-producing ability of the microorganisms. This was evaluated based on the abundance of pyruvate oxidase and NADH-dependent flavin reductase-associated gene families. The microbial species contributing to these gene families were subsequently identified and consolidated for analysis and were visualized using a horizontal lollipop chart. This was done to determine the organisms that have the ability to produce H₂O₂, as well as to determine which microorganisms have the highest hydrogen peroxide-producing ability in the community.

Immune-related functional potential within the microbial community was assessed by identifying pathways associated with host-microbe immune interactions from the unstratified pathway abundance table. A curated set of keyword descriptors was used to filter pathways relevant to immune modulation. To infer potential microbial contributors, species-level associations were derived for each immune-related pathway, identifying taxa contributing to the observed functional potential. Finally, immune-associated pathways were classified into two broad functional categories based on their biological roles: pro-inflammatory pathways, associated with host immune activation and tissue degradation, and homeostatic/structural pathways, associated with baseline microbial metabolism or less directly inflammatory functions. The results of this analysis were visualized using a dot plot.

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RESULTS

Taxonomic Profiling

Post MetaPhlAn Taxonomic Profiling

Following taxonomic profiling of the unmapped reads using MetaPhlAn v4.1.0, the percentages of classified and unclassified reads per sample, together with their mean and standard deviation values, were calculated and consolidated as shown in Table 1.

Table 1. Percentage of classified and unclassified reads per sample

	9228	9229	9230	9231	9233	9234	9235	9236	9237	9238	x̄	SD
Classified Reads (%)	68	77	87	75	65	64	77	77	64	66	72	± 8
Unclassified Reads (%)	32	23	13	25	35	36	23	23	36	34	28	± 8
Total	100	100	100	100	100	100	100	100	100	100	100	

As shown in the table, the percentage of classified reads was consistently higher than the percentage of unclassified reads across all samples. Classified reads ranged from 64% to 87%, with the lowest percentage belonging to samples 9234 and 9237 and the highest percentage belonging to sample 9230. Conversely, the percentage of unclassified reads was consistently lower across samples, ranging from 13% in sample 9230 to 36% in samples 9234 and 9237. The mean percentage of classified reads across all samples was 72% with a standard deviation of $\pm 8\%$, whereas unclassified reads had a mean percentage of 28% with a standard deviation of $\pm 8\%$. 38

After determining the proportion of classified and unclassified reads per sample, the analysis was subsequently focused on the classified reads, particularly on the number of unique taxa identified per sample across different taxonomic levels. These values, together with their mean and standard deviation, are presented in Table 2.

Table 2. Number of unique taxa identified per sample across taxonomic levels

	9228	9229	9230	9231	9233	9234	9235	9236	9237	9238	x̄	SD
Domain	1	1	1	1	1	1	1	1	1	1	1	± 0
Phylum	13	13	13	12	11	11	13	11	12	10	12	± 1
Class	56	51	52	54	47	40	55	51	49	51	51	± 5
Order	67	58	62	65	57	50	66	61	59	62	61	± 5
Family	83	70	75	81	69	60	77	73	71	76	74	± 7
Genus	161	138	132	153	119	117	154	137	149	143	140	± 15
Species	388	301	266	297	302	284	379	306	390	337	325	± 46
Strain	409	313	276	308	323	301	396	323	408	351	341	± 48

As shown in Table 2, across all samples, the number of unique taxa identified generally increases from higher to lower taxonomic resolution. Only one domain was identified across all samples, with a mean and standard deviation of 1 ± 0 , while progressively larger numbers of unique taxa were identified at the phylum, class, order, family, genus, species, and strain levels.

At the phylum level, the number of unique taxa detected per sample ranged from

10 in sample 9238 to 13 in samples 9228, 9229, 9230, and 9235, with a mean and standard deviation of 12 ± 1 . A similar pattern was also observed in class, order, and family level resolution with relatively low standard deviation values (i.e., ± 5 , ± 5 , and ± 7 , respectively).

In contrast, greater variability between samples was observed at the lower taxonomic levels. At the genus level, the number of unique taxa identified ranged from 117 in sample 9234 to 161 in sample 9228, with a mean and standard deviation of 140.39 ± 15 . A similar pattern was observed at both the species and strain levels. Species-level taxa ranged from 266 in sample 9230 to 390 in sample 9237, with a mean and standard deviation of 325 ± 46 , while strain-level taxa ranged from 276 in sample 9230 to 409 in sample 9228, with a mean and standard deviation of 341 ± 48 .

Table 3 summarizes the combined number of unique taxa identified across all samples at each taxonomic level. As shown, a similar increasing trend was also observed in this table. As taxonomic resolution becomes finer, the number of unique taxa becomes higher, from only one domain-level taxon detected, increasing to 15 phyla, 69 classes, 83 orders, 102 families, 230 genera, 575 species, and 604 strains.

Table 3. Total number of unique taxa identified across samples per taxonomic level

Taxonomic level	Total
Domain	1
Phylum	15
Class	69
Order	83
Family	102
Genus	230
Species	575
Strain	604

Identification of the Core Oral Microbiome

The core oral microbiome was used in order to determine the microbial composition of the community by identifying which taxa are most abundant and present in the majority of the samples. The analysis was done at both genus and species levels using a combined prevalence and abundance threshold defined at $\geq 80\%$ and $\geq 0.01\%$, respectively.

A total of 32 taxa were identified as core for both genus and species levels. Only the top 15 taxa were shown in both levels, and the remaining 17 were pooled and designated as “Others”. The results of this analysis are shown in Figure 1A and 1B, which show the core oral microbiome at the genus and species level, respectively, and the distribution of the core per individual sample.

Figure 1. Core oral microbiome and its distribution across samples at the (A) genus and (B) species levels

(A)

(B) 42

At the genus level (Figure 1A), *Neisseria* and *Prevotella* showed the highest relative abundance across the majority of the samples. *Neisseria* was particularly abundant in samples 9228, 9230, and 9233 and was notably lowest in sample 9231, while *Prevotella* had the highest relative abundance in sample 9231. The relative abundances of *Prevotella* were consistently high across samples, with no samples

showing notably low abundances. Other core genera detected include *Poryphromonas*, *Rothia*, *Fusobacterium*, *Alloprevotella*, *Veillonella*, *Streptococcus*, *Capnocytophaga*, *Actinomyces*, *Haemophilus*, *Aggregatibacter*, and *Tannerella*, with relative abundances varying between samples.

Candidatus Saccharibacteria unclassified at the genus level was also identified as one of the core and was present in the majority of the samples, albeit in lower abundances. It had the highest abundance in samples 9231, 9233, and 9236. In addition, an unclassified or provisional genus-level taxon designated as GGB1144 was also detected in some of the samples with varying relative abundances. It had the highest abundance in sample 9237. The proportion of “Others” had a noticeably high relative abundance across samples, with sample 9231 having the highest abundance.

At the species level (Figure 1B), *Neisseria subflava*, *Prevotella melaninogenica*, and *Neisseria sicca* had the highest relative abundance observed across the majority of the samples, although their relative abundances vary. *Neisseria subflava* was particularly abundant in samples 9228, 9233, 9234, and 9237, while it was notably lowest to the point of being unnoticeable in sample 9231. *Prevotella melaninogenica* showed the highest abundance in samples 9235 and 9238 and was notably lowest, almost unnoticeable in sample 9230. *Neisseria sicca* was abundant in 9230, 9233, and 9236, and was particularly lowest, almost unnoticeable in sample 9237. Additional core species identified include *Rothia mucilaginosa*, *Porphyromonas bobii*, 43 *Fusobacterium periodonticum*, *Porphyromonas gingivalis*, *Prevotella intermedia*, *Fusobacterium nucleatum*, *Neisseria elongata*, *Porphyromonas endodontalis*, *Alloprevotella tannerae*, and *Prevotella pallens*.

Two unclassified *Candidatus Saccharibacteria* species bearing the provisional name TM7 phylum sp oral taxon 352 and *Nanosynsacchari* sp TM7 ANC 38 39 G1 1, respectively, were also among the core species identified and were present in some of the samples, although their relative abundances were notably lower.

TM7 phylum sp oral taxon 352 had a noticeable relative abundance in samples 9228, 9229, 9233, 9235, 9236, 9237, and 9238, with a high relative abundance in sample 9236. It was notably lowest in sample 9234 and was absent in samples 9230 and 9231. *Candidatus Nanosynsacchari* sp TM7 ANC 38 39 G1 1 was noticeably abundant in samples 9229, 9230, 9231, 9233, and 9238, with a notably high relative abundance in sample 9231. It was lowest at sample 9228, almost unnoticeable, and was absent in samples 9234, 9236, and 9237. The “Others” category remained a

considerable component in most samples, with the highest relative abundance of almost 50% in sample 9231.

Functional Profiling

Most Abundant Pathways in the Community

This analysis involves the identification of the most abundant pathway in the community. The analysis was done by calculating the mean abundance of each pathway across samples and sorting them in descending order. The top 15 pathways with the highest mean abundances were selected. These pathways are shown in Figure 2.

Figure 2. Top 15 metabolic pathways across samples and their mean abundance in counts per million (CPM)

As shown in Figure 2, all pathways show a completeness score of ≥ 0.8 , indicative of the green color of the lollipop lines. Peptidoglycan maturation (meso-diaminopimelate-containing) was observed to be the most abundant pathway with a mean CPM score of 927.59, considerably higher than the other pathways. The second and third most abundant pathways were the superpathway of adenosine nucleotides de novo biosynthesis I (624.2 CPM) and queuosine biosynthesis I de novo (609.94 CPM), respectively. Folate transformations II (588.64 CPM), CDP-diacylglycerol biosynthesis II (574.42 CPM), and the superpathway of adenosine nucleotides de novo biosynthesis II (573.65 CPM) followed closely. The remaining pathways including peptidoglycan biosynthesis III mycobacteria (566.24 CPM), CDP-diacylglycerol biosynthesis I (565.06 CPM), cis-vaccenate biosynthesis (559.69 CPM), guanosine ribonucleotides de novo biosynthesis (550.45 CPM), inosine 5'-phosphate degradation (550.3 CPM), peptidoglycan biosynthesis I meso-diaminopimelate containing (547.83 CPM), gondoate biosynthesis anaerobic (546.18 CPM), UDP-N-acetylmuramoyl-pentapeptide biosynthesis II lysine-containing (545.8 CPM), and chorismate biosynthesis from 3-dehydroquinate (545.37 CPM), showed relatively similar mean abundances ranging from approximately 545 to 567 CPM.

Species Contribution per Pathway

The species contributions to each of the top 15 pathways were subsequently determined and are presented in Figure 3. The figure shows a heatmap of the pathways along the vertical axis, mapped against the contributing species along the horizontal axis, with cell values representing the strength of their contribution transformed in $\log_{10}(\text{CPM}+1)$.

Figure 3. Contribution of each species in log transformed counts per million + 1 values (CPM + 1) for each of the top 15 pathways

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Across all pathways, the unclassified species group and *Neisseria flavescens* consistently exhibited the highest contribution values, represented by the darkest purple shading in the heatmap. This pattern was most pronounced in peptidoglycan maturation (meso-diaminopimelate-containing), where the unclassified group and *N. flavescens* recorded the highest contributions of 158 and 143, respectively. They are then followed by *Neisseria perflava* (96), *Neisseria mucosa* (90), and *Neisseria subflava* (88). Other *Neisseria* species, including *N. mucosa*, *N. perflava*, and *N. subflava*, also showed moderate to high contributions across most pathways. In contrast, both *Fusobacterium periodonticum* and *Rothia mucilaginosa* showed absent or negligible contributions in several pathways, as indicated by white cells in the heatmap. Similarly, cis-vaccenate biosynthesis had no detectable contributions from *Aggregatibacter segnis*, *F. periodonticum*, *Haemophilus parainfluenzae*, and *R. mucilaginosa*. *Streptococcus mitis* showed notably low contribution to inosine 5'-phosphate degradation, recorded as the lowest non-zero (1) value among all species-pathway combinations. The heatmap revealed that *Neisseria* species and the unclassified group were the primary contributors across the majority of the top metabolic pathways identified in the community.

Analysis of Oral Related Functions

Metabolism of Carbohydrate and pH Balance

The community was assessed based on its ability to metabolize carbohydrates and to balance the pH of the oral cavity as a result of carbohydrate fermentation. The carbohydrate fermentation pathways identified, as well as the species contributing to 47 those pathways, are shown in Figure 4, while the acid-alkali balance profile of the detected microbial species is shown in Figure 5.

Figure 4. Acid-fermentation pathways present in the community and their species contribution

Figure 5. Acid-alkali balance of the detected microbes showing net acidifiers to the left and net alkalizers to the right.

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As shown in Figure 4, pyruvate fermentation to acetate and (S)-lactate I and pyruvate fermentation to acetate and lactate II were the top two carbohydrate fermentation pathways in the community with the broadest distribution of microorganisms. The two fermentation pathways were present in almost all microorganisms shown in the figure, except for *Haemophilus parainfluenzae* and *Prevotella melaninogenica*. The contributions of each microorganism in the two pathways were observed to be in varying degrees, with the highest contribution observed in “unclassified” microorganisms. This is followed by *Rothia mucilaginosa*, *Neisseria flavescens*, and *Neisseria mucosa*. Aside from its contributing microorganisms, the two pathways were also found to be partially complete, as indicated by the orange color box on the further left side of the rows corresponding to the two pathways.

Following the top two pathways, glycolysis III (from glucose) and glycolysis IV were also among the top pathways with the broadest distribution of microorganisms. The two pathways were found in most of the microorganisms except for *Morococcus cerebrosus*, *Neisseria mucosa*, *Neisseria flavescens*, *Rothia mucilaginosa*, and *Actinomyces* sp ICM47, with the latter only absent in the glycolysis IV pathway. Similar to the previously discussed pathways, these two pathways were also highly abundant in “unclassified” microorganisms, with a much higher degree of contribution indicated by the slightly darker red color of the two boxes as compared to the top two pathways. Following this “unclassified” microorganism is *Prevotella melaninonegnica*, which has

a similar degree of contribution in both of the pathways. Glycolysis III (from glucose) was observed to be complete, while glycolysis IV was identified to be incomplete. Glycolysis III (from glucose) was the only pathway identified to be complete among all of the carbon fermentation pathways present in the community. 49

Aside from the top pathways, there were also pathways that were identified to be in low abundance, with only one contributing species. These pathways are the superpathway of glycolysis, pyruvate dehydrogenase, TCA, and glyoxylate bypass, and plant cytosolic glycolysis + PDH + TCA. Both of these pathways were identified to be present only in “unclassified” microorganisms with weak contribution scores, as indicated by the faint red, almost white color of the boxes.

It is important to note as well that the “unclassified” microorganisms had the highest abundance and contributing scores on all the carbohydrate fermentation pathways identified, suggesting that the community is driven by oral microorganisms that have not yet been classified.

On the other hand, Figure 5 shows the acid-alkali balance profile of the microbial community. As shown in the figure, the results of this analysis showed that all the top 20 identified microorganisms exhibited negative balance scores which indicates that the community is predominantly driven by acidogenic pathways instead of alkalizing pathways. The majority of species demonstrated extremely negative results approaching -1. Among the dominant species identified which shows net acidification included *Neisseria flavescens*, *Neisseria mucosa*, *Fusobacterium nucleatum*, *Haemophilus parainfluenzae*, *Streptococcus mitis*, and *Streptococcus oralis*. The analysis found no species with positive balance results indicative of alkalizing metabolic pathways. The “unclassified” group was also able to be detected in this analysis with a relatively high acidogenic potential.

Pathogen Protection

The community was also assessed based on its ability to protect the oral cavity against invading pathogens through its capacity to produce hydrogen peroxide (H_2O_2).

The microorganisms and their hydrogen peroxide-producing capacity are shown in Figure 6. asdasaadsasdadadadadasdaasdasdasdasdasdadadadsadadadadadadadada

Figure 6. Detected species and their hydrogen peroxide production capacity measured in counts per million based on the presence of pyruvate oxidase and NADH dependent flavin reductase gene families

As shown in the figure, a total of 15 microbial species within the community were identified to possess hydrogen peroxide (H₂O₂)-producing capacity based on pyruvate oxidase and NADH-dependent flavin reductase gene family abundance. Among these, *Streptococcus mitis* exhibited the highest predicted H₂O₂-producing capacity, with a total mean gene family abundance of approximately 2.856 CPM, more than twice that of the succeeding species. 51

This was followed by *Streptococcus pneumoniae* (1.632 CPM), while *Streptococcus cristatus* ranked third with a total mean abundance of 1.057 CPM. *Streptococcus* sp A12 also demonstrated comparatively high abundance (1.024 CPM), closely following *Streptococcus cristatus*. The “unclassified” microorganism group was also identified as one of the hydrogen peroxide-producing microbes, albeit with a relatively lower total mean abundance at 0.299 CPM, ranking second to last of the 15 species.

Other taxa identified to have hydrogen peroxide production capacity include *Streptococcus oralis* (0.997 CPM), *Streptococcus sanguinis* (0.849 CPM), *Streptococcus gordonii* (0.840 CPM), *Streptococcus pseudopneumoniae* (0.816 CPM), *Streptococcus australis* (0.571 CPM), *Streptococcus parasanguinis* (0.549 CPM), *Streptococcus infantis* (0.444 CPM), *Streptococcus salivarius* (0.350 CPM), *Streptococcus anginosus* (0.312 CPM), and *Neisseria cinerea* (0.273 CPM).

Immune System Related Functions

Finally, the community was assessed based on the ability of the microorganisms to cause immune-related responses. The results of this analysis are shown in Figure 7.

Figure 7. Immune-modulating pathways detected and the species contributing to those pathways

As shown in the figure, two immune-modulating pathways were identified within the microbial community: gondoate biosynthesis (anaerobic) and the superpathway of (Kdo)2-lipid A biosynthesis. Based on functional annotation, gondoate biosynthesis (anaerobic) was classified as homeostatic/structural, whereas the superpathway of (Kdo)2-lipid A biosynthesis was classified as pro-inflammatory due to its established role in lipopolysaccharide (LPS) formation and immune activation. A total of 12 microbial contributors, including an unclassified taxonomic group, were associated with these pathways. For the homeostatic/structural pathway, all 12 taxa contributed to functional abundance, with the unclassified group exhibiting the highest relative abundance, as indicated by the larger dot size in the figure. This was followed by *Neisseria flavescens*, *Neisseria subflava*, and *Neisseria mucosa*. 53 In contrast, the pro-inflammatory superpathway of (Kdo)2-lipid A biosynthesis was predominantly associated with the unclassified microbial group, which exhibited relatively low abundance, while no substantial contributions from classified species were detected.

DISCUSSION

The taxonomic profiling of the unmapped reads from the whole genome sequence data of the Batak indigenous Filipino group revealed a 72% mean percentage of classified reads across all 10 samples, indicating that the majority, or almost three-quarters, of the microbial genetic material embedded in the human contaminome of the indigenous group could be assigned to known or partially characterized microbial taxa. This high classification rate reflects the improved capacity of MetaPhlAn 4 compared to its predecessors, as this version uses an expanded database containing over one million prokaryotic reference genomes, genomes from cultured known microorganisms, and metagenome-assembled genomes, microbial genomes from uncultured and previously unknown taxa reconstructed using environmental reads, enabling it to classify organisms even if a sequenced isolate is absent (Blanco-Míguez et al. 2023).

Moreover, this mean classification rate is expected and consistent with many metagenomic studies of complex microbial communities. Previous studies reported classification rates ranging from approximately 17-50% in complex microbial communities (Edwin et al. 2025), while oral microbiome studies have reported classification rates ranging from 67.86%, when using a genome database derived only from cultured organisms, to as high as 88.29%, when using metagenome-assembled genome databases (Zhu et al. 2021). This suggests that the acquired mean classification rate is not just accidental but supported by previous studies. Furthermore, the results of this analysis validate the contaminome-based approach as a reliable and efficient culture-independent method for oral microbiome profiling. 55

The observed interindividual variability in classification rates between samples is also not accidental but carries a deeper biological meaning. This between-sample variability, which ranges from 64% in samples 9234 and 9237 up until 87% in samples 9230, reflects the inherent heterogeneity of the oral microbiome across individuals. Each person possesses a distinct oral microbiome that is shaped by a combination of dietary habits, immune history, oral hygiene practices, and environmental influences (Hall et al. 2017). In the context of the Batak group, despite belonging to one cultural group, each individual still possesses their unique microbiome signatures. Samples with

a higher classification rate might suggest that the greater proportion of their oral microbes comes from microorganisms that are well represented in global reference databases, whereas samples with lower classification rates might suggest a larger proportion of undocumented microbial diversity specific to the Batak oral environment. The 28% mean percentage of unclassified reads across 10 samples should also not be ignored. This percentage is expected, as complex microbial communities such as the oral microbiome contain a large proportion of still unclassified or unnamed taxa. In fact, recent studies showed that only 57% of oral bacterial species are officially named, and around 30% still remain uncultivated (Zhu et al. 2021). Furthermore, as explained earlier, the presence of these unclassified reads may also suggest microbial taxa that are highly specific to the group and have never been classified. As Chrisman et al. (2022) established, the human contaminome refers to unmapped reads from whole genome sequencing data that carry genuine biological information, sometimes exceeding the boundaries of current reference databases. In the context of the indigenous Batak group, where little to no studies about their oral microbiome have been published, the existence of a substantial portion of unclassified reads is highly expected and may contain (1) microbial species that are endemic to or enriched in 56 Philippine indigenous oral communities, (2) novel strains of known species with sufficient genomic divergence to evade marker gene matching, and (3) elements such as bacteriophages and archaea that fall outside the bacterial scope of the database. Further studies on the Batak oral microbiome are therefore recommended to uncover the identity of these unclassified reads.

Focusing again on the classified reads, the taxonomic profiling was able to identify one microbial domain, 15 phyla, 69 classes, 83 orders, 102 families, 230 genera, 575 species, and 604 strains. The progressive increase in the number of unique taxa from higher to lower taxonomic levels reflects the hierarchical nature of taxonomic classification, where organisms become increasingly specific and differentiated at finer taxonomic resolutions. The identification of one microbial domain, specifically Bacteria, across all samples suggests that the oral microbial community of the Batak is largely composed of bacterial members. This reflects the dominance of bacterial representatives in the oral microbiome (Lu et al. 2019; Caselli et al. 2020). Moreover, the detection of only one domain may also reflect the limitations of the reference database used and the detection limits of the other microbial components of the oral microbial community. Other than Bacteria, the oral microbial community also harbors

fungi, protozoan, archaeal, and viral representatives, although their abundance is significantly lower compared to Bacteria, limiting their detection in many taxonomic classification tools (Achtman and Zhou 2020; de Cena et al. 2022; Rajasekaran et al. 2024; Crouch et al. 2024). Going down to the species and strain levels, the detection of 575 species and 604 strains is also consistent with the established composition of the oral microbiome reported in numerous studies, including those catalogued by the Human Oral Microbiome Database (HOMD) (Chen et al. 2010). These findings suggest that despite being a distinct cultural group, the oral microbial composition of the Batak 57 appears broadly consistent with patterns reported in other human oral microbiome studies, particularly at higher taxonomic levels.

The greater inter-individual variability between samples at the species (± 46) and strain level (± 48) compared to the phylum (± 1) and class levels (± 5), reflects a well-established principle in microbiome science, which states that the higher order community structure of microbiomes is relatively conserved across individuals and populations, while the species and strains populating that framework differ substantially between hosts. This pattern has been consistently observed in large-scale human microbiome studies and is thought to reflect the combined influence of various factors such as host genetics, diet, and immune history on the composition of the microbiome at finer resolutions (Turnbaugh et al. 2007). For the Batak, this variability in species and strains may additionally be shaped by the diversity of microhabitats within the oral cavity itself, including the tongue dorsum, buccal mucosa, and supragingival and subgingival plaque, each of which harbors distinct microbial niches among individuals, especially in the absence of standardized site-specific sampling, such as in this study, where sampling was performed only through saliva.

Moving to the specific microbial composition of the Batak community, the identification of the 32 core oral microbial taxa at the genus and species level using a combined prevalence threshold of $\geq 80\%$ and a relative abundance threshold of $\geq 0.01\%$ establishes a well-defined baseline of the microbial community of the Batak. At the genus level, the identification of *Neisseria* and *Prevotella* as the dominant genera across the majority of the samples is ecologically coherent and consistent with global literature on human oral microbiome and the specific lifestyle context of the Batak Filipinos. A comprehensive large-scale metagenomic analysis of over 4,000 oral samples identified 58 *Neisseria* as one of the most prevalent oral microbial genera across diverse human populations (Zhu et al. 2021). Moreover, in the specific lifestyle context of Batak as

hunter-gatherers, the identification of *Neisseria* as one of the prevalent genera is consistent with the study of Lassalle et al. (2018), who demonstrated that oral microbiomes of hunter-gatherer and semi-foraging populations in the Philippines tend to be enriched in commensal *Neisseria* as compared to the populations with more Westernized diets. This also reflects the dietary composition of the group with limited consumption of refined carbohydrates and processed foods (Il and Eder 2006; Negrito, Palawan Batak in Philippines people group profile | Joshua Project nd; Peoples of the Philippines: Batak - National Commission for Culture and the Arts nd), allowing commensal aerobic taxa such as *Neisseria* to be favored in the oral cavity rather than anaerobic-fermentative microorganisms.

On the other hand, the notable high abundance of *Prevotella* in sample 9231, while *Neisseria* had a low abundance, reflects the inverse ecological relationship of these two genera in the oral cavity. This ratio is influenced by dietary substrates, oxygen levels, and salivary flow rate in the mouth (Könönen et al. 2022), which determines whether the oral cavity will be dominated by *Prevotella* or *Neisseria*. The observed inter-individual variation in the *Prevotella*-*Neisseria* ratio reflects the sample-specific ecological dynamics operating within the Batak oral community.

At the species level, *Neisseria sicca* and *Neisseria subflava* were among the core oral microbiome identified in the Batak oral community. The identification of these species is consistent with the literature on the core oral microbiome of humans (Aas et al. 2005; Dewhirst et al. 2010). In addition, *Prevotella melaninogenica* was also among the species identified as core. This *Prevotella* species is a well-characterized anaerobic commensal of the oral mucosa that is consistently present from early life onward in healthy individuals (Könönen et al. 2022). The dominance and consistency of *Neisseria* taxa across taxonomic resolutions and across samples suggests that *Neisseria* and its associated species may play a central role in shaping the observed community profile.

Another notable finding of this study is the identification of *Porphyromonas* and *Fusobacterium* as core members of the Batak oral cavity. Specifically, the study was able to identify at the species level *Porphyromonas gingivalis*, *Porphyromonas endodontalis*, *Fusobacterium periodonticum*, and *Fusobacterium nucleatum*. In the literature, these microbial species are classically described as pathogens associated with periodontal diseases and endodontic infections (Bedran et al. 2012; Ciani et al. 2024). However, their presence as core members of the Batak oral microbiome requires careful

interpretation and should not be immediately considered as pathogenic and indicative of disease. Examination of their distribution shows that these species are present at low to moderate abundances across samples, a pattern consistent with a non-pathogenic or context-dependent state rather than pathogenic within a balanced microbial community. This observation is also consistent with the literature on *P. gingivalis*, *P. endodontalis*, *F. periodonticum*, and *F. nucleatum*, which describes them as opportunistic pathogens (Bostanci and Belibasakis 2012; Chen et al. 2022). This means that, as opportunistic pathogens, they do not inherently cause disease but can become pathogenic under specific conditions, such as host immune dysregulation and microbial dysbiosis (Bruslind 2019). However, interpreting low-abundance opportunistic pathogens as purely commensal should also be approached with caution, as certain species, such as *P. gingivalis*, have the capacity to contribute to disease even at low abundance (Hajishengallis and Lamont 2014). This is consistent with the keystone pathogen hypothesis, which proposes that specific low-abundance pathogens can drive inflammatory disease by remodeling the microbiota into a dysbiotic state rather than relying on high abundance alone (Hajishengallis et al. 2012).

Aside from opportunistic pathogens, another scientifically significant finding of this study is the identification of two *Candidatus Saccharibacteria* (TM7) taxa: TM7 phylum sp. oral taxon 352 and *Candidatus Nanosynsacchari* sp. TM7 ANC 38 39 G1 1 as one the core oral microbiome of the Batak. These microorganisms belong to the Candidate Phyla Radiation (CPR), an evolutionary lineage of bacteria whose members are mostly uncultivated and known only through metagenomics and single-cell sequencing (Ji et al. 2022). The detection of these two microbes demonstrates the scientific value of culture-independent analysis in detecting uncultured microorganisms. If culture-dependent analysis were employed, these taxa would have been absent in the results.

Ecologically, *Saccharibacteria* are obligate episymbionts, meaning that they are microorganisms that live on the surface of other microorganisms. In the case of *Saccharibacteria*, their most common host are Actinomyces-related species. Since they are obligate episymbionts, these microorganisms derive their nutrients solely from their bacterial host (Bor et al. 2019). Their reduced genomes and ultra-small size, characteristics of microorganisms belonging to the CPR division (Ji et al. 2022), are an important adaptation for this type of lifestyle. Despite their parasitic dependencies, evidence shows that they also participate in oral health-related activities. A study by

Chipashvili et al. (2021) demonstrates that they have the ability to suppress gingival inflammation and bone loss in experimental models, particularly in mice. Providing more evidence of the presence of *Saccharibacteria* in the oral cavity is a landmark study by Naud et al. (2023), which confirmed that *Saccharibacteria* is a genuine member of the human oral microbiota. This study provides evidence that the presence of this microbial group is biologically significant rather than being an artifact. Furthermore, their identification in the oral cavity of the Batak indigenous Filipino group also expands the documented geographic distribution of oral *Saccharibacteria* in Southeast Asia.

Functional profiling of the Batak oral microbial community, on the other hand, revealed that the peptidoglycan maturation (meso-diaminopimelate-containing) pathway is the most abundant metabolic pathway identified across the community. The identification of this pathway as the most abundant suggests that the community is likely engaged in active cellular growth and structural maintenance. This is supported by the role of peptidoglycan as the principal polymer that makes up most bacterial cell walls and is responsible for maintaining structural integrity and osmotic pressure. Furthermore, the identification of the meso-diaminopimelate-containing variant, commonly associated with many gram-negative bacteria (Galinier et al. 2023), is consistent with the core oral microbiome of the Batak identified in this study, which is dominated by gram-negative genera, such as *Neisseria*, *Prevotella*, and *Fusobacterium*. Thus, the functional signal observed appears to be coherent with the underlying taxonomic composition of the community.

At the species level, the “unclassified” microorganism group and *Neisseria flavescens* were among the dominant microbial contributors to this pathway. The identification of these two microorganisms reinforces two critical points. First, the “unclassified” being the dominant group suggests that they are not just merely a taxonomic placeholder, rather they are a functionally significant contributor to the core metabolic pathway in the community. This reinforces the significance of culture-independent analysis in uncovering the identity of these “unclassified” microorganisms. Second, the dominance of *N. flavescens* and other *Neisseria* species (*N. perflava* at 96 CPM, *N. mucosa* at 90 CPM, and *N. subflava* at 88 CPM) supports the inference that this genus is the primary metabolic driver of the Batak oral microbiota. This is consistent with its identification as the dominant genera in the core oral microbiome. Aside from being in a potential state of active cellular growth and maintenance,

the community is also likely in a state of active metabolic self-sufficiency as evidenced by the identification of pathways related to de novo biosynthesis, particularly superpathway of adenosine nucleotides de novo biosynthesis I (624.2 CPM), queuosine biosynthesis I de novo (609.94 CPM), superpathway of adenosine nucleotides de novo biosynthesis II (573.65 CPM), and guanosine ribonucleotides de novo biosynthesis (550.45 CPM). The presence and relative abundance of these pathways suggests that the community have the genetic potential to produce the necessary molecular components, particularly those required for DNA and RNA synthesis, from scratch rather than relying solely on premade ones. Given that de novo biosynthesis is energetically expensive, the oral environment likely have adequate materials and resources to support these biological processes. The simultaneous detection of inosine 5'-phosphate degradation (550.3 CPM) further indicates that both biosynthetic and degradative pathways are represented within the community. This may reflect a balance between anabolic and catabolic processes, which is a common feature of complex and dynamic microbial ecosystems. 63

In addition, the detection of folate transformations II (588.64 CPM) as one of the most abundant pathway, suggests that the community also has the potential to generate key intermediates required for amino acid biosynthesis and other cellular processes. Similarly, the detection of fatty acids and membrane lipid biosynthesis pathways including CDP-diacylglycerol biosynthesis I and II (565.06 and 574.42 CPM), cis-vaccenate biosynthesis (559.69 CPM), and gondoate biosynthesis anaerobic (546.18 CPM) may also suggest that the community has the potential for membrane lipid production.

Supporting these interpretations, pathway completeness scores of ≥ 0.8 were observed across all 15 pathways, indicating that the enzymatic machinery required for these pathways is largely present and fully represented in the community. Together, the detection of these pathways suggests that the Batak oral microbial community is functionally capable of supporting active growth and maintenance, biosynthetic activity, and metabolic flexibility. However, it is important to note that these inferences are based on pathway abundance and therefore reflect functional potential rather than direct evidence of in situ metabolic activity.

Apart from the most abundant pathways, the community was also assessed for its potential ability to conduct three oral cavity related functions specifically metabolism of carbohydrates and pH balance, protection against pathogens in the oral

cavity, and immune related functions. Analysis of the potential ability of the community to metabolized carbohydrates revealed that the Batak oral microbial community displayed a dominance of carbohydrate fermentation, with pyruvate fermentation and core glycolytic pathways observed as the most dominant pathways. This suggests that the community may be functionally bias towards carbohydrate fermentation and its 64 associated pathways. This bias towards acidogenic orientation is supported by literature which demonstrates that pyruvate functions is a key metabolic intermediate in oral bacteria which directs carbon metabolism toward organic acid production that decreases oral pH levels (Takahashi and Nyvad 2011; Yu et al. 2026). This is further reinforced by the results of the acid-alkali balance analysis where all top 20 microorganisms identified exhibited negative balance scores without any species showing the ability to produce net alkalizing effects.

The community was also assessed for its potential ability to protect the oral cavity against harmful and invading pathogens. The results of this analysis showed that the Batak oral cavity has an oral microbiome capable of producing hydrogen peroxide (H_2O_2), a natural antimicrobial compound released by some bacteria to destroy or disable dangerous microbes (Williams et al. 2025). The study was able to identify 15 bacterial species as H_2O_2 -producers based on the presence and abundance of the genes required for this function particularly pyruvate oxidase and NADH-dependent flavin reductase. *Streptococcus mitis* showed the highest gene family abundance with a mean value of 2.856 CPM which is twice the mean abundance of the second H_2O_2 -producing microorganism, *Streptococcus pneumoniae* (1.632 CPM). This finding is consistent with recent laboratory studies of *S. mitis* as a strong H_2O_2 producer preventing biofilm formation by caries pathogens such as *Streptococcus mutans* (Williams et al. 2025). The *mitis* group of streptococci which includes *S. mitis*, *S. oralis*, *S. sanguinis*, and *S. gordonii*, all of which have been identified in the Batak oral microbial community, has been established in the literature as a major H_2O_2 -producing commensal that exists in the human oral cavity (Redanz et al. 2018). Oral *Streptococcus* species produce H_2O_2 as a byproduct of aerobic metabolism through the enzyme pyruvate oxidase and this compound has been shown to inhibit oral pathogens. The discovery of multiple 65 microorganisms that have the potential of producing H_2O_2 suggest that this community have the genetic machinery to be able to protect itself from dangerous microbial pathogens.

It is important to note, however, that the “unclassified” microorganisms group

produced the second lowest H₂O₂ output of 0.299 CPM. Given that the “unclassified” group dominated most of the metabolic pathways in this study, their limited contribution in this category may reflect gaps in the current genomic databases regarding novel or understudied species. It is possible that some of these “unclassified” microbes might possess H₂O₂-producing genes which current reference databases fail to identify. This represents a common limitation in culture-independent studies that investigate indigenous populations with limited knowledge of their oral microbiomes (Lamont et al. 2018).

Finally, assessment of the immune-modulating potential of the community revealed that two functionally distinct pathways contribute to immune modulation. These pathways are the gondoate biosynthesis (anaerobic) and the superpathway of (Kdo)₂-lipid A biosynthesis with the former being the most dominant as evidenced by the participation of most of the species identified to contribute to immune modulation. The detection of the gondoate biosynthesis pathway as the more prevalent of the two pathways suggests that the community is may be functionally skewed towards metabolic processes associated with structural maintenance and microbial stability, rather than pro-inflammatory signaling pathways.

Gondoate biosynthesis pathway is considered as homeostatic or structural because it preserves the standard structural integrity and stability of bacterial cell membranes (Zhang et al. 2024). The pathway was observed to be contributed by 12 66 different taxa which showed “unclassified” group as the main component. This is then followed by *Neisseria flavescens*, *Neisseria subflava*, and *Neisseria mucosa*. In the context of the Batak indigenous group who live in close relationship with nature and have restricted access to contemporary medical facilities, the detection of a microbiome that has the potential to maintain immune balance, may represent microbial functional patterns associated with immune modulation potential (Marcano-Ruiz et al. 2023; Dobon et al. 2023).

Contrasting this finding, is the detection of the superpathway of (Kdo)₂-lipid A biosynthesis which is involved in the production of lipid A, a structural element of lipopolysaccharide (LPS). Lipid A acts as an immunostimulatory molecule which the host immune system recognizes to trigger innate immune responses. Host immune activation occurs when LPS recognition through Toll-like receptor 4 (TLR4) initiates inflammatory signaling cascades (Marcano et al. 2021; Zenobia and Darveau 2022). The Batak oral microbiome showed low abundance of this pathway and is mainly linked

to the “unclassified” microorganisms group suggesting that the functional potential for LPS-associated immunostimulatory activity is present but not strongly represented within the characterized members of the community.

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CONCLUSIONS

Overall, the findings of this study provide the first culture-independent, contaminome-based characterization of the Batak oral microbiota, establishing a scientifically grounded microbial baseline for a previously uncharacterized indigenous Filipino group. More specifically, the present study was able to characterize 72% of the human contaminome data into known or partially identified microbial taxa, reinforcing the reliability of the human contaminome approach in culture-independent oral microbiome research. The study was also able to establish that the Batak oral microbiota is taxonomically diverse, as evidenced by the increasing number of unique taxa identified as taxonomic resolution becomes finer across multiple taxonomic levels. In addition, the study was also able to identify 32 core oral taxa dominated by commensal *Neisseria* and *Prevotella* along with their associated species, as well as opportunistic pathogens present at non-pathogenic abundances, and *Candidatus Saccharibacteria*, a group known only through metagenomic and single-cell sequencing. Functional profiling indicates that the community possesses the genetic potential to support processes related to cellular growth and maintenance, biosynthesis, carbohydrate fermentation, antimicrobial compound production through hydrogen peroxide production, and immune-related interactions. However, these inferences reflect predicted functional capacity based on pathway abundance rather than direct evidence of in situ activity. In addition to these taxonomic and functional findings, the study also established that interindividual variation was most pronounced at the species and genus levels, while overall community-level structure remained consistent with global oral microbiome patterns.

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RECOMMENDATIONS

The present study was able to characterize oral microbes in only one domain.

This is a direct limitation of the reference database used in characterizing the microbial species. For future studies, it is recommended that an expanded database which includes other microbial domain as well as acellular microorganisms is used in order to capture the real composition of the oral microbiota of the Batak community.

A notable proportion of unclassified reads as well as the dominance of “unclassified” microorganisms across most pathways was also documented. Future studies focusing on these uncharacterized microorganisms are therefore recommended to uncover their taxonomic identity.

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